

# Wen Fan

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## RESEARCH SUMMARY

Ph.D. researcher in software verification and static analysis. I combine separation-logic-based verification and LLM-assisted specification generation for heap-manipulating programs, aiming to apply formal methods to security-critical and distributed systems.

## EDUCATION

**Ph.D., Electrical and Computer Engineering, Purdue University**

West Lafayette, Indiana, USA

- Advisor: Prof. Jenna DiVincenzo

*Aug. 2022 – Aug. 2028 (Expected)*

**B.S., Computer Science, University of Science and Technology of China (USTC)**

Hefei, Anhui, China

- GPA: 3.75/4.3

*Sept. 2018 – Jun. 2022*

## PUBLICATIONS

- **Wen Fan**, Minh Tran, Sanya Dod, Xin Hu, Marilyn Rego, Danning Xie, Jenna DiVincenzo, Lin Tan  
An Empirical Study of LLM-Generated Specifications for VeriFast  
Under submission to **ISSTA 2026**
- Marilyn Rego\*, **Wen Fan\***, Xin Hu, Sanya Dod, Zhaorui Ni, Danning Xie, Jenna DiVincenzo, Lin Tan  
Evaluating the Ability of Large Language Models to Generate Verifiable Specifications in VeriFast  
**FORGE @ ICSE 2025** [[paper](#)][[code](#)]

## RESEARCH PROJECTS

**Evaluating LLMs on Generating Specifications for VeriFast**

Purdue University

Advisor: Prof. Jenna DiVincenzo

*Sept. 2024 – Now*

- Built a benchmark based on 60 VeriFast programs
- Developed automated Python pipeline to evaluate 10 LLMs on 2700+ outputs
- Fixed and categorized 1648 VeriFast errors from 450 sampled output functions
- Identified only 31.7% verification success; Over 87% functional behavior preserved; 71% of errors caused by tool-specific knowledge gaps

**Optimizing Gradual Program Verifier with Parallel Symbolic Execution**

Purdue University

Advisor: Prof. Jenna DiVincenzo

*Jun. – Aug. 2024*

- Discovered the performance bottleneck in sequential execution by analyzing a program of 571 branches
- Designed and implemented parallel execution of branches, reducing static verification time by 41%

**Consensus Protocol Optimization under Different Fault Models**

Purdue University

Advisor: Prof. Yongle Zhang

*May. 2023 – Mar. 2024*

- Investigated 40+ consensus protocols in different fault models
- Conducted a survey about 5+ synthesis techniques on distributed systems/protocols
- Identified 8+ examples of Paxos optimization under a different fault model

**Static Analysis For Failure Detection in Distributed Systems**

Purdue University

Advisor: Prof. Yongle Zhang

*Jun. 2022 – Apr. 2023*

- Examined 100+ failure cases on Apache Jira to characterize cascading failures
- Implemented a static analyzer using Soot to recognize retry pattern in distributed systems (e.g., Hadoop)

## AWARDS

Huawei Scholarship

*Nov. 2020*

Huaxia Talent Program, USTC

*Jun. 2022*

## SKILLS

**Programming Languages:** C, C++, Python, Java, Scala, Go, Rust, OCaml, Verilog, SQL, MATLAB

**Verification/Analysis:** VeriFast, Dafny, Soot, LLVM

**Systems/Tools:** Linux, Git, Docker, Makefile, LaTeX, Django, Wireshark

**Languages:** English (fluent), Chinese (native)